

EE 210L: Digital Circuits and Logic Design Lab

Lab 10: First FPGA Design

Name: _____ Date: _____

Partner(s): _____ Section: _____

Board: PYNQ-Z2, part xc7z020clg400-1. FPGA I/O runs at 3.3 V. Never connect a 5 V signal to a board pin.

Part 5.2: Majority Voter on Hardware



Figure 1: Top-level module source.



Figure 2: Your XDC constraints file.

sw[0]	sw[1]	btn[0]	LD0 observed	Expected M	Match?
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

Part 5.3: Resource Utilization

Resource	Used	Available	% Utilization
LUTs		53,200	
Flip-flops		106,400	
Bonded IOBs			

The majority voter took three gates on the breadboard. How many LUTs here, and why is a LUT able to absorb the whole function?

Why does K-map minimization save nothing inside a single LUT?

Part 5.4: Two-Bit Comparator on Hardware

a[1:0]	b[1:0]	LD0 (gt)	LD1 (eq)	LD2 (lt)	Exactly one lit?
00	00				
00	01				
00	10				
00	11				
01	00				
01	01				
01	10				
01	11				
10	00				
10	01				
10	10				
10	11				
11	00				
11	01				
11	10				
11	11				

Part 5.5: Break a Constraint on Purpose

Pin changed: _____ moved from _____ to _____

What did the board do?

The Verilog never changed. What does that tell you about the role of the XDC file?

Part 5.6: An Unconstrained Port

Copy the error or critical warning verbatim.

At which stage did it appear: synthesis or implementation?

Why does an unconstrained port fail there rather than at synthesis?

Discussion and Conclusions

Address at least: what a bitstream actually is, how the FPGA flow compares with wiring a breadboard, and which stage of the flow you would check first for a language error versus a pin error.
